ENGINEERING & ELECTRICAL SCHEMATIC DRAWING (DRAUGHTSMANSHIP GUIDE)

Standard Symbols, Control Circuit Schematics, Power Wiring & Industrial Blueprints

SECTION 1: STANDARD ELECTRICAL SYMBOLS (IEC 60617 STANDARD)

KEY SYMBOLS & NOTATIONS FOR WIRING SCHEMATICS:
[NO] Normally Open Contact (Contactor Auxiliary 13-14, Start Push Button)
[NC] Normally Closed Contact (Overload Trip 95-96, Stop Push Button 1-2, Interlock 21-22)
[COIL] Contactor / Relay Operating Solenoid Coil (Designated A1 - A2)
[OLR] Thermal Overload Relay Bimetallic Trip Element (Designated 95-96 Trip, 97-98 Alarm)
[CB] Circuit Breaker with Thermal & Magnetic Trip Elements (1P, 2P, 3P, 4P)
[EARTH] Protective Earth symbol (3 horizontal tapering lines connected to vertical stem)

SECTION 2: DIRECT-ON-LINE (DOL) MOTOR STARTER SCHEMATIC

CONTROL WIRING STEP-BY-STEP TRACE (230V CONTROL):
1. Live Supply -> Control MCB (6A) -> Thermal Overload NC (95-96).
2. Overload NC (96) -> Stop Push Button NC (1-2).
3. Stop PB (2) -> Start Push Button NO (3-4) in parallel with Contactor Auxiliary NO (13-14) [LATCH].
4. Start PB (4) / Contactor NO (14) -> Contactor Coil Terminal A1.
5. Contactor Coil Terminal A2 -> Neutral (N) return busbar.

SECTION 3: FORWARD - REVERSE MOTOR CONTROL INTERLOCKING

ESSENTIAL ELECTRICAL INTERLOCKING RULES:
* Interlocking prevents KM1 (Forward) and KM2 (Reverse) from energizing at the same time.
* Feed KM1 coil through KM2 NC (21-22) contact.
* Feed KM2 coil through KM1 NC (21-22) contact.
* Swapping any two supply phases (L1 and L3) reverses rotating magnetic field in 3-phase induction motor.